



Technical Document

E-Space Solutions (Pvt.) Ltd



K. Vishnuwaran

ESOFT METRO CAMPUS

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Technical Documentation for E-Space Solution

Overview

The E-Space Solution is a database management system developed to assist E-Space Solutions (Pvt.) Ltd. in their Mars colonization project, starting in 2040. The system manages critical data related to colonists, their dependents, E-Jets, trips, housing, and job assignments on Mars. The solution ensures data consistency, accuracy, and ease of retrieval for decision-making purposes, and will be expanded to include reporting features in future development phases.

System Requirements

- Database Management System (DBMS): Microsoft SQL Server
- Operating System: Windows 10 or higher
- Hardware Requirements: Minimum 8 GB RAM, 500 GB storage
- Software Requirements:
 - SQL Server Management Studio (SSMS)
 - NET Framework (for additional tools and integration, if needed)
 - Visual Studio (for integration with the E-Space system application)

User Access Levels

- **System Administrator:** Manages the overall system, data integrity, and security, and generates critical reports.
- **Data Entry Operators:** Input new data (e.g., colonists, trips, dependents) and update existing records.
- **Pilots:** Access trip-related data and passenger details.
- **Colony Superintendent:** Reviews reports on housing, job allocations, and colonist details.

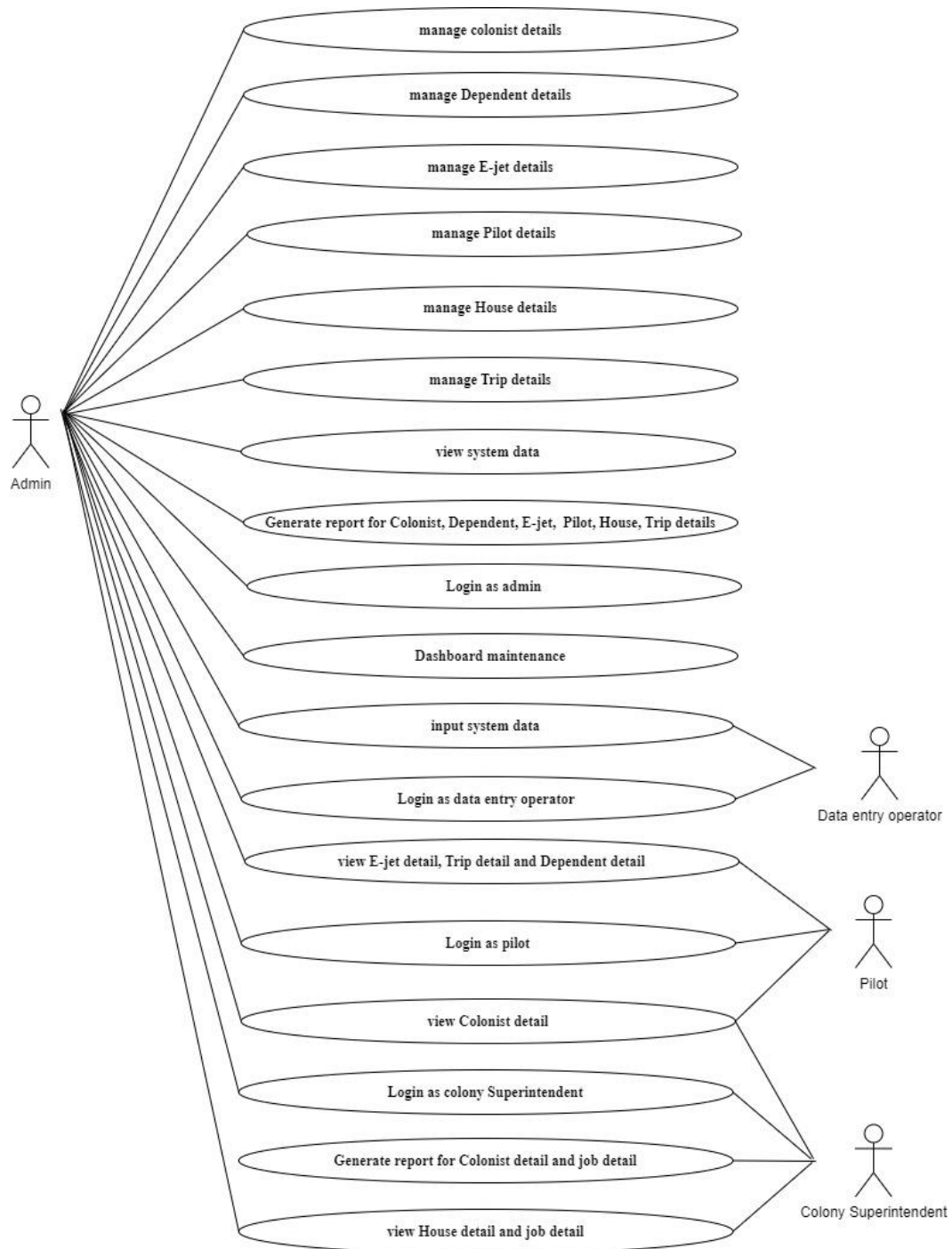
Database Operations

- **Data Input:** Data entry operators use input forms to add colonist details, dependent data, and E-Jet trip information.
- **Search and Retrieval:** System users can search for colonists, trips, or E-Jets using the system's search functionalities, which utilize indexed fields such as Mars_Coloni_id, Jet unique Code, or Trip id.
- **Data Validation:** The system implements validation constraints, such as unique Mars_Coloni_id, and ensures referential integrity across all tables (e.g., linking colonists to dependents).
- **Reports:** Future implementation of reporting will include detailed colonist reports, trip schedules, and housing assignments.

Error Handling

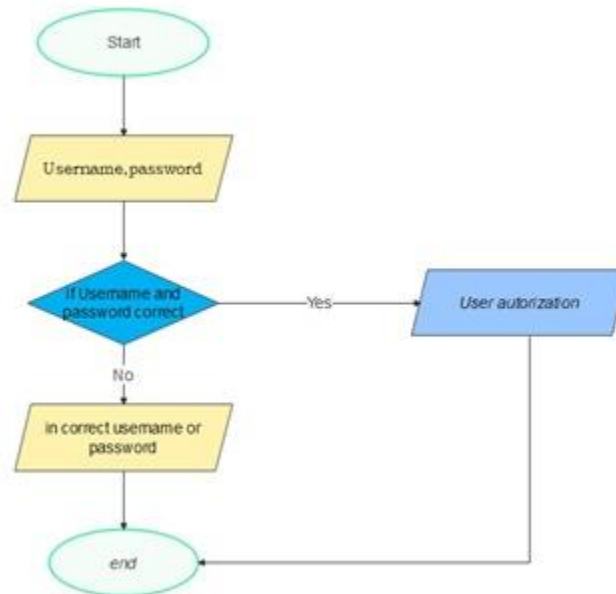
- **User Errors:** Detailed error messages guide users when invalid data is entered, such as duplicate Mars Colonization ID or missing required fields.
- **System Errors:** All system-level errors, such as database connection failures, are logged and displayed to administrators with relevant debug information.
- **Transaction Rollback:** In case of database errors, transactions are rolled back to ensure data integrity is maintained.

Use case Diagram for E Space Solution

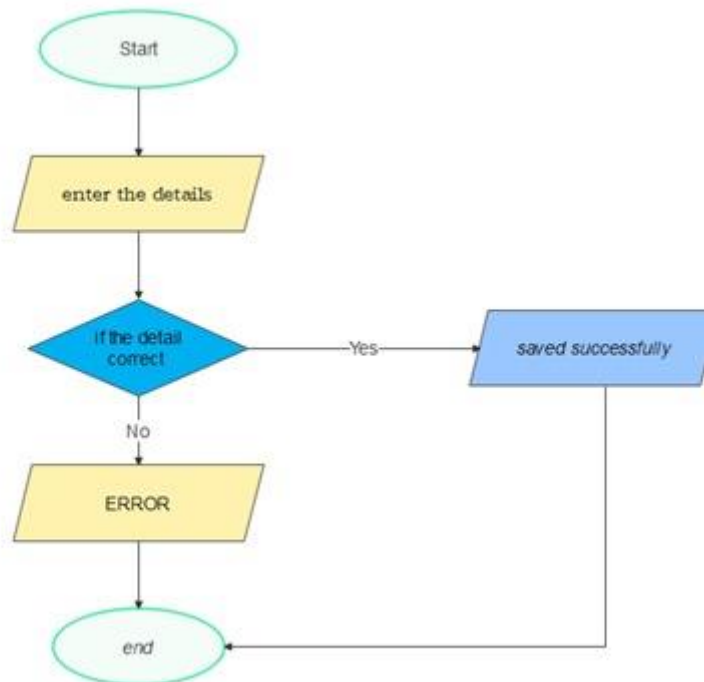


Flow Chart for Database Functionalities

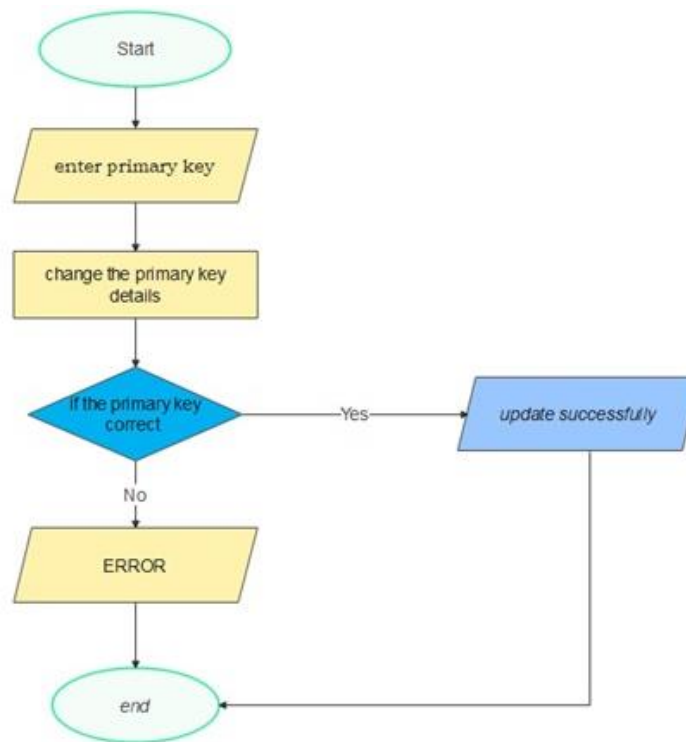
1. Login



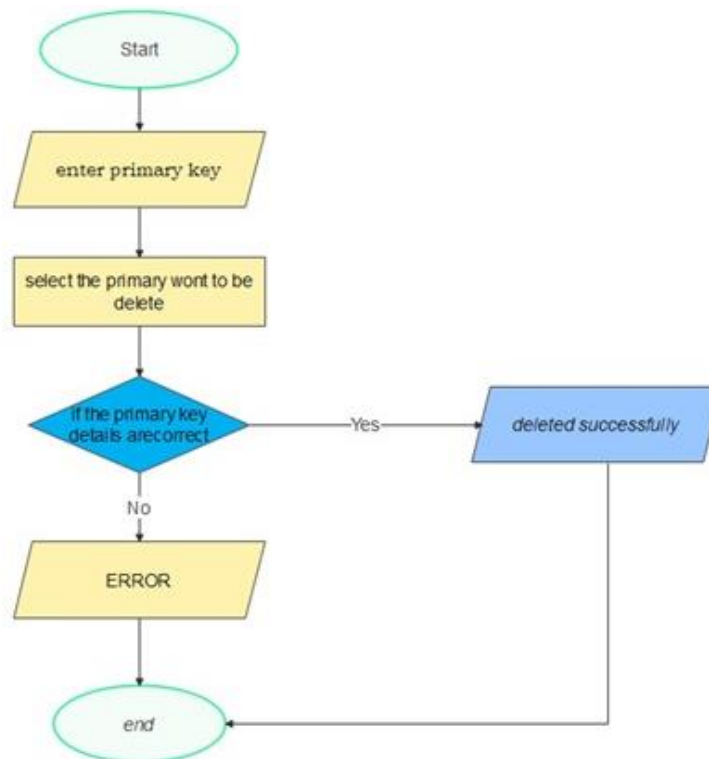
2. Insert



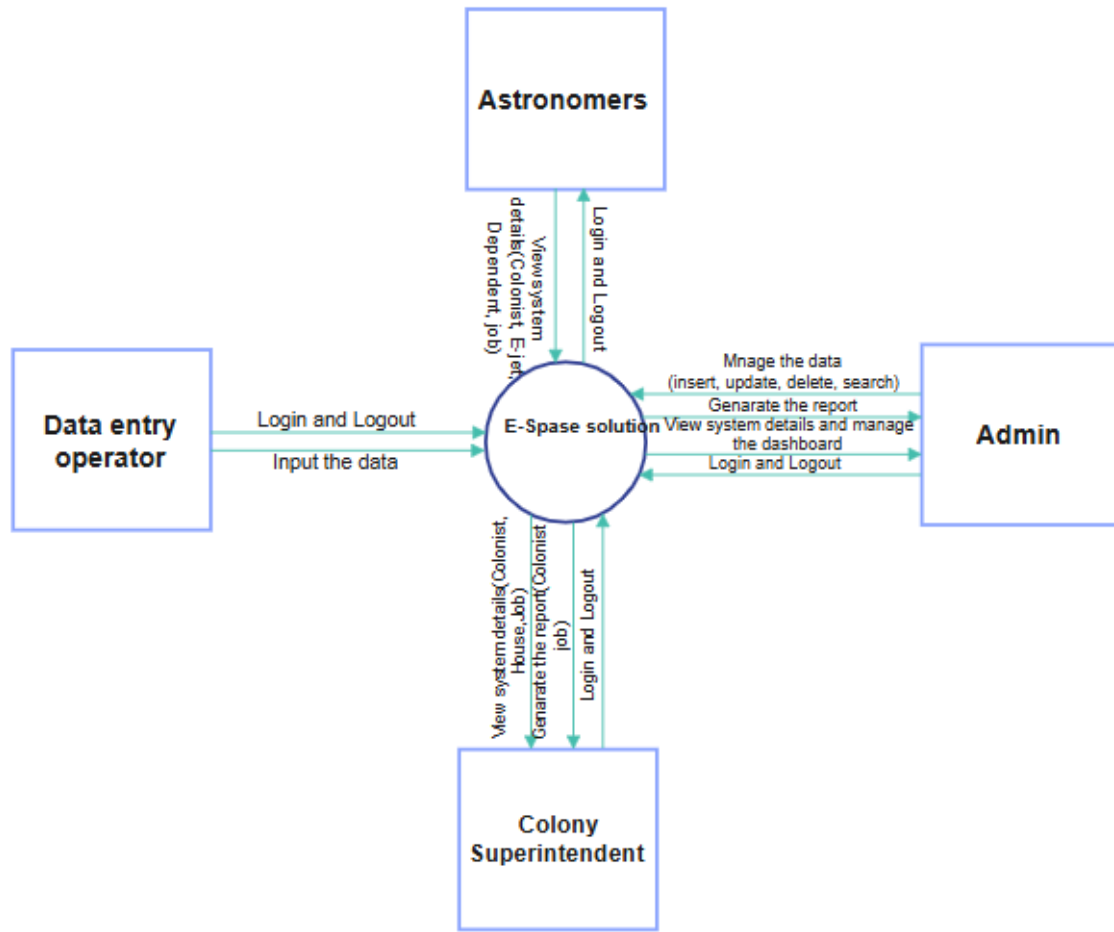
3. Update



4. Delete

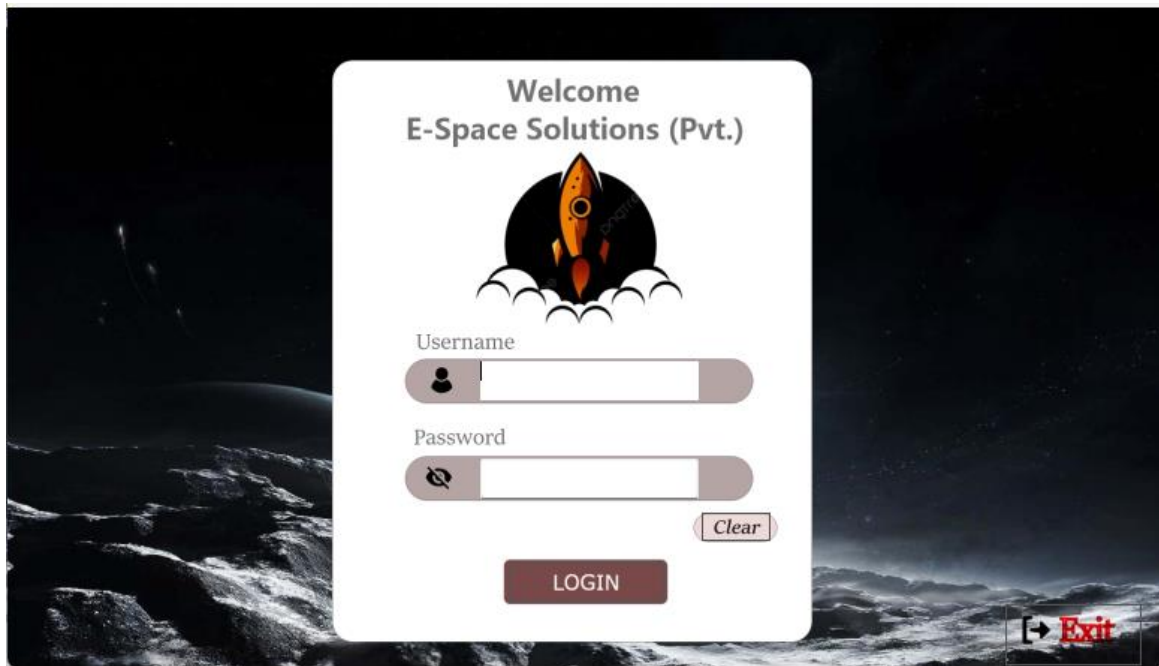


Dataflow Diagram for E Space Solution



System Interface with code

- Login interface



```
1  using System;
2  using System.Collections.Generic;
3  using System.ComponentModel;
4  using System.Data;
5  using System.Drawing;
6  using System.Linq;
7  using System.Text;
8  using System.Threading.Tasks;
9  using System.Windows.Forms;
10 using System.Data.SqlClient;
11
12 namespace E_SPACE
13 {
14     public partial class Form1 : Form
15     {
16         SqlConnection con = new SqlConnection(@"Data Source=DESKTOP-KNPLAV8\SQLEXPRESS;Initial Catalog=Own_Espace;Integrated Security=True");
17         public Form1()
18         {
19             InitializeComponent();
20         }
21
22         private void button2_Click(object sender, EventArgs e)
23         {
24         }
25
26         private void button1_Click(object sender, EventArgs e)
27         {
28         }
29
30         private void button1_Click_1(object sender, EventArgs e)
31         {
32             String UserName = textBox1.Text;
33             String Password = textBox2.Text;
34             if (ValidateLogin(UserName, Password))
35             {
36                 MessageBox.Show("Login Succesfully");
37             }
38         }
39     }
40 }
```

```

48     }
49 }
50
51 1 reference
52 private void button2_Click_1(object sender, EventArgs e)
53 {
54     textBox1.Clear();
55     textBox2.Clear();
56 }
57 1 reference
58 private bool ValidateLogin(string username, string password)
59 {
60     if ((username == "vishnu" && password == "0326"))
61     {
62         return true;
63     }
64     return false;
65 }
66 1 reference
67 private void button3_Click(object sender, EventArgs e)
68 {
69     DialogResult result = MessageBox.Show("Do you want to exit this page?", "Confirm Exit", MessageBoxButtons.YesNo, MessageBoxIcon.Question);
70
71     if (result == DialogResult.Yes)
72     {
73         Application.Exit();
74     }
75 }
76
77
78
79
80
81 0 references
82 private void textBox2_TextChanged(object sender, EventArgs e)
83 {
84 }
85
86

```

Other Functionalities interfaces with code

E-Space Solution (Put.)

House Details

Colony_lot_number

Num_of_rooms

Square_feet

CLEAR

INSERT **UPDATE** **DELETE** **BACK**

```

1  using System;
2  using System.Collections.Generic;
3  using System.ComponentModel;
4  using System.Data;
5  using System.Drawing;
6  using System.Linq;
7  using System.Text;
8  using System.Threading.Tasks;
9  using System.Windows.Forms;
10 using System.Data.SqlClient;
11
12 namespace E_SPACE
13 {
14     5 references
15     public partial class Form3 : Form
16     {
17         1 reference
18         SqlConnection con = new SqlConnection(@"Data Source=DESKTOP-KNPLAV8\SQLEXPRESS;Initial Catalog=Own_Espace;Integrated Security=True");
19         public Form3()
20         {
21             InitializeComponent();
22         }
23
24         1 reference
25         private void button1_Click(object sender, EventArgs e)
26         {
27             Form2 x = new Form2();
28             x.Show();
29             this.Hide();
30         }
31
32         1 reference
33         private void Form3_Load(object sender, EventArgs e)
34         {
35         }
36
37         1 reference
38         private void button2_Click(object sender, EventArgs e)
39         {
40             if (textBox1.Text == "" || textBox2.Text == "" || textBox3.Text == "")
41             {
42                 MessageBox.Show("fillout the empty fields");
43             }
44         }
45     }
46 }

```

```

47     }
48 }
49
50 1 reference
51 private void button2_Click(object sender, EventArgs e)
52 {
53     if (textBox1.Text == "" || textBox2.Text == "" || textBox3.Text == "")
54     {
55         MessageBox.Show("fillout the empty fields");
56     }
57     else
58     {
59         con.Open();
60         SqlCommand cmd = new SqlCommand("insert into house (Colony_lot_number,Num_of_rooms,Square_feet) values ('" + textBox1.Text + "
61         cmd.ExecuteNonQuery();
62         MessageBox.Show("saved success");
63         con.Close();
64     }
65 }
66
67 1 reference
68 private void button3_Click(object sender, EventArgs e)
69 {
70     if (DialogResult.Yes == MessageBox.Show("Do you want to update the data ?", "conformation", MessageBoxButtons.YesNo, MessageBoxIcon.
71     {
72         con.Open();
73         SqlCommand cmd = new SqlCommand("Update house set Num_of_rooms='" + textBox2.Text + "',Square_feet='" + textBox3.Text + "' where
74         cmd.ExecuteNonQuery();
75         MessageBox.Show("Update successful");
76         //clearmethod();
77         //gridviewupdate();
78
79         con.Close();
80     }
81 }

```

```

    }
}

1 reference
private void button4_Click(object sender, EventArgs e)
{
    if (DialogResult.Yes == MessageBox.Show("Do You Want Delete ?", "Confirmation", MessageBoxButtons.YesNo, MessageBoxIcon.Warning))
    {
        SqlCommand cmd = new SqlCommand("DELETE FROM house where Colony_lot_number like '" + textBox1.Text + "'", con);
        con.Open();
        cmd.ExecuteNonQuery();
        MessageBox.Show(" house Details Deleted Successfull", "Details Deleted", MessageBoxButtons.OKCancel, MessageBoxIcon.Information)

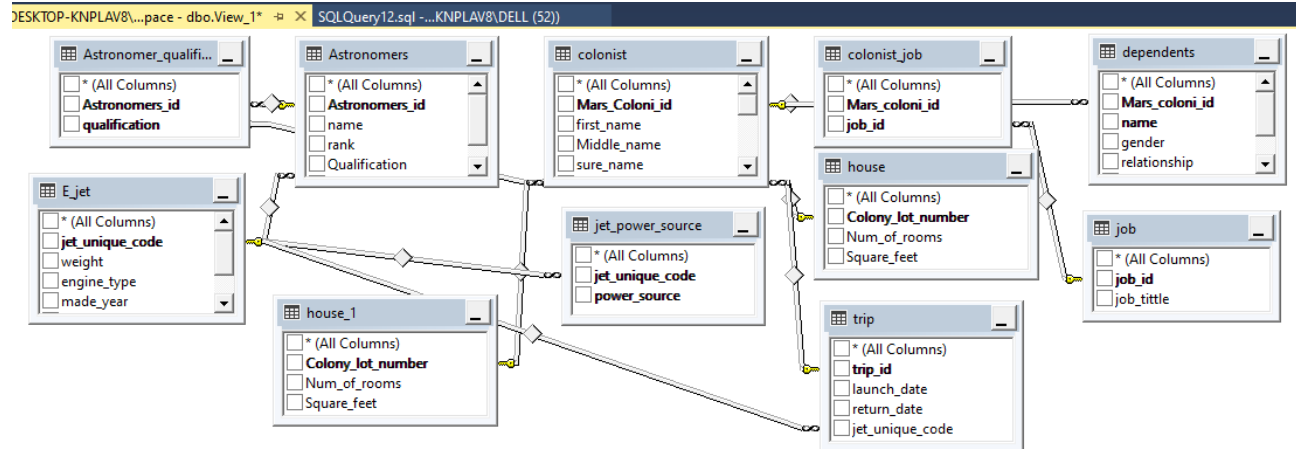
        con.Close();
    }
}

1 reference
private void button5_Click(object sender, EventArgs e)
{
    textBox1.Clear();
    textBox2.Clear();
    textBox3.Clear();
    MessageBox.Show("clear successfully");
}

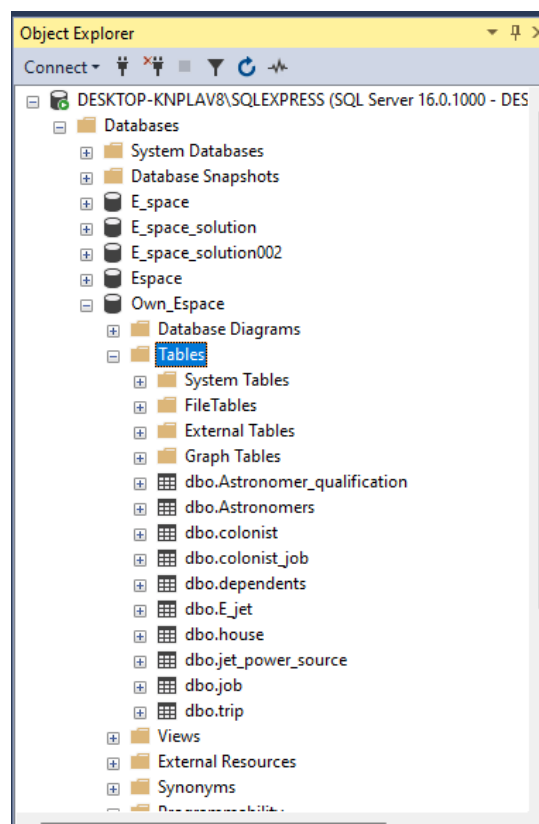
1 reference
private void button6_Click(object sender, EventArgs e)
{
    SqlCommand cmd = new SqlCommand("select* from house where Colony_lot_number like'" + textBox4.Text + "'", con);
    con.Open();
    SqlDataReader sdr = cmd.ExecuteReader();
    if (sdr.Read())
    {
        textBox2.Text = sdr["Num_of_rooms"].ToString();
        textBox3.Text = sdr["Square_feet"].ToString();
        MessageBox.Show("data found");
    }
    else

```

Design view of tables



Database data tables



A watercolor splash in shades of blue and teal serves as a background for the 'Thank You' text.

Thank You

